```sql
select * from `retail.sales` LIMIT 1000;


--Total cost of each product
select InvoiceNo, Quantity, UnitPrice, (Quantity*UnitPrice) as amount
from `retail.sales`;

with bills as (
 select InvoiceNo, Quantity, UnitPrice, (Quantity*UnitPrice) as amount
from `retail.sales`
)
select InvoiceNo, SUM(amount) as total from bills
group by InvoiceNo


-- groupby customer

select s.CustomerID,
DATE(MAX(s.InvoiceDate)) as last_purchase_date,
DATE(MIN(s.InvoiceDate)) as first_purchase_date,
COUNT(DISTINCT s.InvoiceNo) as num_purchase,
SUM(b.total) as monetary
from `retail.sales` as s
left join `retail.bills` as b
on s.InvoiceNo = b.InvoiceNo
group by CustomerID;


with cte as (
 select s.CustomerID,
DATE(MAX(s.InvoiceDate)) as last_purchase_date,
DATE(MIN(s.InvoiceDate)) as first_purchase_date,
COUNT(DISTINCT s.InvoiceNo) as num_purchase,
SUM(b.total) as monetary
from `retail.sales` as s
left join `retail.bills` as b
```

```sql
on s.InvoiceNo = b.InvoiceNo
group by CustomerID
)
select *,
DATE_DIFF(reference_date, last_purchase_date, DAY) as recency,
num_purchase / month_cust as frequency from
(select *, MAX(cte.last_purchase_date) OVER() as reference_date,
DATE_DIFF(cte.last_purchase_date, cte.first_purchase_date, month) + 1 as month_cust
from cte
)


-- percentiles

select a.*,
b.percentiles[offset(20)] as m20,
b.percentiles[offset(40)] as m40,
b.percentiles[offset(60)] as m60,
b.percentiles[offset(80)] as m80,
b.percentiles[offset(100)] as m100,

c.percentiles[offset(20)] as f20,
c.percentiles[offset(40)] as f40,
c.percentiles[offset(60)] as f60,
c.percentiles[offset(80)] as f80,
c.percentiles[offset(100)] as f100,

d.percentiles[offset(20)] as r20,
d.percentiles[offset(40)] as r40,
d.percentiles[offset(60)] as r60,
d.percentiles[offset(80)] as r80,
d.percentiles[offset(100)] as r100
from `retail.RFM` a,
(select approx_quantiles(monetary, 100) percentiles from `retail.RFM`) b,
(select approx_quantiles(frequency, 100) percentiles from `retail.RFM`) c,
(select approx_quantiles(recency, 100) percentiles from `retail.RFM`) d
```

```
ORDER BY CustomerID


select CustomerID,
m_score, f_score, r_score, recency, frequency, monetary,
cast(round((f_score+m_score)/2, 0) as INT64) as fm_score
from (
 select *,
 case when monetary<=m20 then 1
       when monetary>m20 and monetary<=m40 then 2
       when monetary>m40 and monetary<=m60 then 3
       when monetary>m60 and monetary<=m80 then 4
       when monetary>m80 and monetary<=m100 then 5
 end as m_score,

 case when frequency<=f20 then 1
       when frequency>f20 and frequency<=f40 then 2
       when frequency>f40 and frequency<=f60 then 3
       when frequency>f60 and frequency<=f80 then 4
       when frequency>f80 and frequency<=f100 then 5
 end as f_score,

 case when recency<=r20 then 5
       when recency>r20 and recency<=r40 then 4
       when recency>r40 and recency<=r60 then 3
       when recency>r60 and recency<=r80 then 2
       when recency>r80 and recency<=r100 then 1
 end as r_score
 from `retail.quantile`
)


-- RFM Segment

select CustomerID,
recency, frequency, monetary, r_score, f_score, m_score, fm_score,
```

```sql
case when (r_score=5 and fm_score=5)
     or (r_score=5 and fm_score=4)
     or (r_score=4 and fm_score=5)
     then 'Champions'
   when (r_score=5 and fm_score=3)
     or (r_score=4 and fm_score=4)
     or (r_score=3 and fm_score=5)
     or (r_score = 3 and fm_score=4)
     then 'Loyal Customers'
  WHEN (r_score = 5 AND fm_score = 2)
       OR (r_score = 4 AND fm_score = 2)
       OR (r_score = 3 AND fm_score = 3)
       OR (r_score = 4 AND fm_score = 3)
   THEN 'Potential Loyalists'
   WHEN r_score = 5 AND fm_score = 1 THEN 'Recent Customers'
   WHEN (r_score = 4 AND fm_score = 1)
       OR (r_score = 3 AND fm_score = 1)
   THEN 'Promising'
   WHEN (r_score = 3 AND fm_score = 2)
       OR (r_score = 2 AND fm_score = 3)
       OR (r_score = 2 AND fm_score = 2)
   THEN 'Customers Needing Attention'
   WHEN r_score = 2 AND fm_score = 1 THEN 'About to Sleep'
   WHEN (r_score = 2 AND fm_score = 5)
       OR (r_score = 2 AND fm_score = 4)
       OR (r_score = 1 AND fm_score = 3)
   THEN 'At Risk'
   WHEN (r_score = 1 AND fm_score = 5)
       OR (r_score = 1 AND fm_score = 4)
   THEN 'Cant Lose Them'
   WHEN r_score = 1 AND fm_score = 2 THEN 'Hibernating'
   WHEN r_score = 1 AND fm_score = 1 THEN 'Lost'
   END AS rfm_segment
 from `retail.score`
 order by CustomerID;
```